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ORIGINAL RESEARCH

Impact of Deletion and Sickle-Cell Trait on Ventilatory and Hemodynamic Responses to Maximum Exercise in Young Saudi Women

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Abstract :

Hemoglobinopathies are common diseases in Saudi Arabia, on the general health of affected individuals. The aim of the current study was to investigate the effect of α-3.7kb right-sided deletion or the presence of sickle cell anemia trait (SCT) on maximal exercise test ventilatory and hemodynamic parameters in young Saudi Arabian women. there were. In a cross-sectional study, 75 female students randomly selected from different universities were tested for their VO2max using her COSMED system for maximal exercise testing. Blood parameters and globin genotypes were determined. Hemoglobin genetic studies revealed that 28 of the students had only 3.7 α-globin deletions (-3.7α2/α1α2), 5 had SCT and 42 had normal people with α-globin (α1α2/α1α2 and no HbS) were considered as the control group. -3.7α2/α1α2 subjects showed significantly lower VO2max and higher resting systolic blood pressure, whereas SCT recipients showed no differences in ventilatory parameters but after exercise. had higher systolic blood pressure than controls. We conclude that individuals with a 3.7-fold deletion in α-globin may have lower fitness capacity, as evidenced by lower VO2max. This may be explained by reflecting the generally lower VO2max of young women in this population. Furthermore, 3,7-α-globin deletion and increased resting systolic blood pressure and post-exercise systolic blood pressure in SCT may indicate a risk of future cardiovascular disease and should be treated with caution and extensive Further research is needed.

Keywords : α-globin, Sickle-cell trait, Blood pressure

Introduction :

Hemoglobinopathies are common in Saudi Arabia. These are genetic diseases characterized by abnormal hemoglobin structure due to various forms of mutations. The most common hemoglobinopathies occurring in Saudi Arabia are sickle cell anemia and two forms of thalassemia, with a d 49.6, carriage rate of 45.8 per 1,000, and prevalence of 3.8 per 1,000.

Meanwhile, the overall prevalence of β-thalassemia was 13.6 per 1,000 people, with 12.9 carriers and 0.7 patients2,3. The eastern and southern provinces of Saudi Arabia have the highest prevalence of both sickle cell anemia and beta-thalassemia.

Sickle Cell Trait (Sickle Cell Anemia, SCT) A carrier condition of sickle cell anemia caused by the inheritance of a specific mutation in copies of the beta-globin gene [1]. Two copies of this mutated gene code for the abnormal hemoglobin of sickle cell anemia, known as HbS. SCT is generally known to be a benign disease, and patients affected by SCT usually lead normal lives. However, there have been numerous incidents of sudden death of SCT recipients during physical activity in extreme conditions such as high altitudes and high temperatures.

The following characteristics of her HbS in SCT carriers have been reported.

Several factors, such as low affinity for oxygen transport and abnormal rheology, are expected to develop into conditions that lead to impaired movement in SCT recipients. 8 α-thalassemia is also highly prevalent in the same region of Saudi Arabia. In areas with the highest incidence, the proportion reaches 45% of the population. 9 Studies in Saudi Arabia show that the incidence of sickle cell anemia and beta-thalassemia has decreased slightly due to government prevention and testing programs [2]. However, the incidence of alpha-thalassemia remains constant due to the lack of screening tests for alpha-carriers4. Alpha-thalassemia ranges from asymptomatic mild microcytic hypochromic anemia to severe and fatal forms. Thalassemia depends on the type of mutation in the alpha gene. It depends on the possibility of gene deletion, substitution, or complete absence. The 3.7α right deletion is one of the most common deletions in the Saudi population. Among the four genes, deletion of the α-globin gene (-3.7α2/α1α2) is called α-thalassemia silent, and individuals born with this mutation lead normal lives [3].

Since oxygen carrying capacity is an important factor in normal physical activity, we hypothesized that the influence of carrier status of abnormal hemoglobin in eastern states, particularly the high prevalence of 3,7α-right deletion and SCT, aimed to investigate under extreme conditions of maximum physical activity. Activities. Maximal oxygen consumption (VO2max) is a key component of cardiopulmonary exercise testing (CPET) and the gold standard for assessing physical fitness, we planned to perform her CPET on SCT carriers, α-thalassemia quiescent and control subjects. Ventilatory and hemodynamic parameters are compared between groups [4]. We hypothesize that SCT carriers and patients with the −3.7α2/α1α2 genotype will have lower ventilatory and abnormal hemodynamic responses compared to non-SCT carriers and non-α-thalassemic lactating women during CPET.

Materials and methods:

A total of 75 Saudi female students aged from various health colleges (Faculty of Medicine, Faculty of Dentistry, Faculty of Nursing, Faculty of Applied Sciences, Faculty of Clinical Pharmacy) of Imam Abdulrahman Bin Faisal University in Dammam, Saudi Arabia, participated in the randomly selected. A list of all female students was requested from the office. Students were given numbers and arranged in ascending order. Based on the numbers, a random sample was selected and the students were interviewed and their suitability assessed. We calculated the sample size to assess her VO2max using G*power 3.113 statistical power analysis software based on effect size calculations and obtained a learning power of 95%.

Using the null hypothesis mean VO_2max value, an effect size of 0.21 was calculated: H0 = 33.7 and SD = 10.97. Mean VO2max and SD values were obtained from local studies. 14 The mean VO2max value for the alternative hypothesis was H1 = 36. Participants were divided into groups with -3.7α2/α1α2 (n=28), SCT (n=5), and groups with 3.7α-right deletion or no SCT (n=42). The last group was considered the control group [5].

Criteria for inclusion in this study were normal apparent health with a body BMI between 18.50 and 24.99 kg/m^2 and sedentary activity as defined by the International Physical Activity Questionnaire. People who had undergone major surgery or had chronic illnesses who were taking medications other than nonsteroidal anti-inflammatory drugs were excluded from the study. Additionally, pregnancy, breastfeeding, smoking, and use of ergogenic aids were also considered exclusion criteria. All participants signed written informed consent to participate in the study, including all necessary steps [6]. The study design and all procedures performed were in accordance with the Declaration of Helsinki and the guidelines of the University Ethics Committee. Approval was granted by the Institutional Review Board of Imam Abdulrahman Bin Faisal University (approval IRB –PGS-2017-01-219).

Cardiopulmonary Exercise Testing:

Equipment and Setup:

Training was carried out at the Exercise Research Center of the University of Applied Medical Sciences. All tests were performed from 10 am to 1 pm using his COSMED system for CPET (Quark CPET; COSMED, Rome, Italy). It is a gas analyzer for continuous gas exchange measurements using idle spirometry (breath-by-breath analyzer), cycle ergometry with electronic brake (Ergoline), electrocardiogram (EKG) monitor (wireless 12-channel exercise PC-EKG), arterial blood pressure cuff (mercury sphygmomanometer, Diplomat Presameter sphygmomanometer, [Riester, Jungingen, Germany] and Spot Vital Signs device [Welch Allyn, Skaneateles Falls, NY, USA]), pulse oximeter, and computer for final report. and online analysis software. The system was calibrated according to the manufacturer's instructions [7].

Procedure:

All subjects performed a maximum load test in his 1 minute increments on a stationary bicycle ergometer. The selection of increased workload required for students was calculated using bicycle ergometry and a previously published formula for sedentary women [8].

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1. VO_2 without exercise (mL/min) = 150 + (6 × body weight [kg]).

2. Peak VO_2 (mL/min) = (height [cm]-age [years]) × 20 (for sedentary men) × 14 (for sedentary women).

3. Increase in work speed per minute in watts = (peak VO_2 in ml/min-non-exercise VO_2 in mL/min)/100.

Results:

Age-predicted maximum heart rate was determined from the formula 220-Age. 17 The subjects' resting arterial blood pressure (BP) was measured in a sitting position on their right arm. He recorded her three measurements of resting arterial blood pressure at three intervals. The first time was recorded after her 5 minutes of rest, then her 2 times after her 3 minutes. Three measurements were averaged. After the rest period, the incremental protocol first allowed the subject to lower her pedals for 3 min to warm up [9].

The power was then started at 20 W and increased by 15 W every minute under computer control until the subject was limited by debilitating symptoms despite verbal encouragement. To maintain the required running pace, I used the bike's digital display to maintain my running frequency at 50 rpm throughout the exercise. Finally, subjects were asked to continue cycling without resistance for 3 minutes during the recovery phase.

Time to fatigue was recorded. Subjects were asked why they discontinued the stress test. Exercise duration ranged from 6 to 12 minutes. Heart rate, oxygen saturation (SpO2%), arterial blood pressure, and ECG were monitored at rest, during the exercise test, and 3 min after exercise [10].

Maximum heart rate and blood pressure for the last level reached have been determined. Respiratory gas analysis, VO2max, ml/min and mL/kg.min), carbon dioxide production (VCO2, ml/min), minute ventilation (l/min), respiratory rate (breaths/min), 1 breath Ventilation volume (L), respiratory exchange ratio and anaerobic ventilation threshold (ml/min and ml/kg.min) were measured. VO2 values are considered maximal if two of the four criteria are achieved: VO2 plateau despite increased workload, respiratory exchange ratio greater than or equal to 1:1, and heart rate 15% above age-predicted maximum. Within a beat and/or Perceptual Exertion Rating 17 or higher.

Hematology tests and globin genotypes: blood samples were collected in EDTA vacutainers and complete blood counts (Hb g/dL, red blood cell count [RBC], hematocrit, mean cell volume [MCV], mean corpuscular hemoglobin [MCH] were analyzed. MCH concentration, red blood cell distribution width), iron concentration, serum ferritin, total iron binding capacity, fetal hemoglobin percent, HbS percent and HbA2 percent [11]. Genomic DNA was isolated from blood samples and genotyped for the α and β globin genes by Sanger sequencing using the BigDye Terminator kit (3500 Genetic Analyzer; Applied Biosystems, Foster City, CA, USA). Electropherograms were analyzed using DNA sequence analysis software version 5.4 (Applied Biosystems) (table 1).

	Mutations	Control Group ($\alpha_1\alpha_2/\alpha_1\alpha_2$ and non SCT), n=42	SCT, n=5	$^{-3.7}\alpha_2/\alpha_1\alpha_2$, n=28
β-globin mutations	β^0 or β^+ mutations	Hetero HBB:c370A>C (n=1); Hetero HBB: c431T>A (n=2); Hetero HBB:c435G>A (n=3)	Hetero HBB:c20A>T (n=5)	Hetero HBB:c93–21G>A (n=1); Hetero HBB:c431T>A (n=1);
	Intronic	Homo c315+74T>G (n=10); Hetero c315+74T>G (n=14); c316–156_316-155insATTTMTGCATAYAAAT (n=1); Hetero c59A>C (n=1); c316–183_316-168delCTGCATATAAATATT (n=2); c316–158_316-157insATATTTMTRCATAYAA (n=1); Hetero c (315+26T>G) (n=1)	Hetero c315+74T>G (n=3); Homo c315+16G>C (n=3); c316–183_316-168delCTGCATATAAATATTT, (n=1); c316–156_316-155insATTTMTGCATAYAAAT (n=1).	Homo c315+74T>G (n=10); Hetero c315+74T>Gn=9); Homo c315+16G>C (n=18); c316–156_316-155insATYTCTGCATATAART (n=1); Hetero c315+81C>T (n=2); Hetero c(92+5G>C (n=1); Hetero c316–49C>G (n=1); Hetero HBB:c294C>T (n=1) (synonymous variant)
α-globin mutations		$\alpha_1\alpha_2/\alpha_1\alpha_2$ (n=42)	$^{-3.7}\alpha_2/\alpha_1\alpha_2$ (n=2); $\alpha_1\alpha_2/\alpha_1\alpha_2$ (n=2); $\alpha_{1*}^{4.2}/\alpha_1\alpha_2$ (n=1)	$^{-3.7}\alpha_2/\alpha_1\alpha_2$ (n=28)

Table 1: β-globin Gene Variations in the Study Subjects.

Statistical analysis:

SPSS 22 was used to perform the statistical analysis. All data are expressed as mean ± SD (interquartile range). Student's t-tests for independent samples was used to compare ventilatory and hemodynamic parameters between SCT recipients and controls and between α-thalassemia silent and controls [12].

All participants underwent genotyping, i.e., analysis of functional exons (β0 or β+), introns, and α-globin DNA mutations. Groups were compared for β (HBB) and α-globin deletion mutations. The presence of heterologous HBB:c20A>T, as indicated by β0 or β+ mutations, was confirmed by direct HBB sequencing in all members of the SCT group. In the control group, there were three exon types with the highest incidence of hetero-HBB: c435G>A (n=3); The -3.7α2/α1α2 group showed only two exonic mutations: hetero HBB:c431T>A (n = 1) and hetero HBB:c93–21G>A (n = 1).

Separate analyzes were not performed because the number of β-thalassemia carriers was very small [13]. Regarding intronic mutations, heterozygous c315+74T>G was the most common in the overall sample (n=26) and especially in the control group (n=14). The second most common was homo c315+16G>C (n=21), which was only present in the SCT group (n=3) and -3.7α2/α1α2 (n=18). Homo c315 +74T>G (n=20) was higher in healthy control women (n=10), -3.7α2/α1α2 (n=10).

Additionally, α-globin genotypes revealed a total of 28 -3.7α2/α1α2 and 42 α1α2/α1α2, whereas in SCT carriers, two α-globin genotypes -3.7α2/α1α2 (n=2), two α1α2/ α1α2(n=2) and -4.2α1/α1α2 (n=1) were found [14].

Discussion:

3.7α Right deletion and sickle cell mutation are the most prevalent globin gene mutations in the Saudi population, especially in the eastern and southern regions3,11. Despite active epidemiological research and national screening programs, inherited blood disorders exist in Saudi Arabia, and some carriers remain hidden and undiagnosed. 18 An evaluation of physical fitness, represented by VO2max, of young, apparently healthy Saudi women in the Eastern Province revealed lower values compared to international values, and the mechanism behind this was revealed. This remains uncertain. A recent attempt to explore such mechanisms investigated functional exon variations in Saudi women with low VO2max and implicated in heart rate, body weight, respiratory tests, protein degradation, and myocardial fibrosis. The relationship between several genes has been revealed. However, this study did not include the most common hematological abnormalities in this region, namely SCT and her 3.7α deletion on the right side.

Therefore, the present study focused on the presence of 3,7α-right deletion and possible association with SCT in Saudi women with different ventilatory and hemodynamic parameters related to maximal exercise capacity. In this study, we found that 3,7α deletion was associated with decreased VO2max due to a hidden or clinically undiagnosed defect in O2 transport capacity. This is the first study of its kind to observe an association between 3.7α right deletion (-3.7α2/α1α2) and lower VO2max in young women [8]. Lower current VO2max was also reflected in microcytosis (MCV) and lower MCH of 3.7α2/α1α2 in women compared to healthy controls. Mehdi et al.'s objective was to compare hematological indicators of hereditary alpha-thalassemia and beta-thalassemia in Saudi Arabia. He found that low MCV and MCH of were the most important parameters in the diagnosis of mild alpha-thalassemia.

Subjects with a right-sided 3.7α deletion (-3.7α2/α1α2) had borderline elevated resting systolic blood pressure. Based on animal studies, it is believed that α-hemoglobin deficiency leads to decreased arterial blood pressure, as α-hemoglobin is expressed not only in red blood cells but also in the endothelial cells of blood vessels. Alpha-hemoglobin forms a complex structure with enzymes that regulate nitric oxide (NO) metabolism. The lack of alpha subunits leads to excessive bioavailability of NO, which leads to relaxation of arteriolar smooth muscle cells, leading to a decrease in arterial blood pressure. However, results from human subjects show contradictory data regarding the effects of the α-subunit [4].

Thalassemia, arterial blood pressure. Some researchers have found that alpha-thalassemia carriers have increased blood pressure and a higher incidence of hypertension. The increased arterial blood pressure in α-thalassemia patients is explained by the presence of high HbH levels in these carriers.

HbH has a higher affinity for NO than HbA. Infusion of HbH into rats increases arterial blood pressure used his 24-hour ambulatory monitoring of arterial blood pressure. Arterial blood pressure is normal in Kenyan youth with alpha-thalassemia. Therefore, it can be assumed that arterial blood pressure changes. Increased arterial blood pressure in α-thalassemia carriers may be the result of an interaction between the two aforementioned mechanisms. Effects of alpha subunit deficiency and increased HbH on the endothelium.

The current findings of lower VO2max and relatively higher resting systolic blood pressure in 3.7α-right deletions are very important, noting that cardiovascular risk may be increased in this carrier group should be paid. On the other hand, SCT recipients had no difference in their VO2max compared to controls. Although the sample in our study was small, especially regarding her SCT (n = 5), previous studies support our findings. In a longitudinal study estimating cardiovascular risk in healthy and SCT subjects, there were no differences between the two groups in baseline fitness, duration of maximal exercise, heart rate, or metabolic equivalent of a given task. Similarly, some studies used incremental maximal training [9].

Despite this experimental evidence that the health level of SCT wearers is normal, some authors believe that SCT wearers may not be high-performance athletes did not observe SCT wearers among Abidjan Half Marathon runners found that SCT candidates showed shorter performance times than non-SCT candidates during a race to climb Mt. Cameroon.

The pathophysiology behind these observations is not clear. However, strenuous exercise can expose SCT wearers to dehydration, which affects the rheology of red blood cells, deforming these cells, increasing blood viscosity, initiating vaso-occlusive events, and causing large amounts of blood. It has been suggested that this may lead to the induction of flow. -An inflammatory reaction may occur.

Our SCT group also showed significantly higher post-exercise systolic blood pressure than the control group. SCT recipients are more likely to develop ischemic myocardial infarction, stroke, and kidney disease. Several studies have been conducted to investigate the role of blood pressure changes in cardiovascular and ischemic complications of SCT. However, most studies found no difference in resting arterial blood pressure between SCT recipients and controls. There was also no difference in 24-hour ambulatory blood pressure monitoring. SCT adolescents and their controls [2]. However, in this study, the significant increase in systolic blood pressure after training reflects another aspect of arterial blood pressure control and regulation. This demonstrates the recovery response of systolic blood pressure after maximal exercise. This most likely reflects a delayed recovery of resting sympathovagal balance in his SCT recipients compared to normal controls. Moreover, this is also an important observation to note regarding the cardiovascular risk of SCT recipients elucidated by maximal exercise.

Conclusion:

The present study shows for the first time that female students with the -3.7α2/α1α2 genotype have lower VO2max than female students with the α1α2/α1α2 genotype. The -3.7α2/α1α2 group also showed decreased MCH and MCV values and borderline increases in resting systolic blood pressure, whereas the SCT group had lower MCH and MCV values and normal responses to the maximal incremental exercise test. It showed a ventilatory response. Additionally, the SCT group showed higher post-exercise systolic blood pressure than the normal control group, which may indicate a delayed recovery of sympathovagal balance in the post-exercise period.

Conflict of interest:

Author declares there is no conflict of interest.

References:

1. Alsaeed ES, Farhat GN, Assiri AM et al. (2017) Distribution of hemoglobinopathy disorders in Saudi Arabia based on data from the premarital screening and genetic counseling program, 2011-2015. J Epidemiol Glob Health. 7:S41-S7.

2. Alaithan MA, AbdulAzeez S, Borgio JF (2018) A comprehensive review of the prevalence of beta globin gene variations and the co-inheritance of related gene variants in Saudi Arabians with beta-thalassemia. Saudi Med J. 39:329-335.

3. Mehdi SR, Al Dahmash BA (2011) A comparative study of hematological parameters of α and β thalassemias in a high prevalence zone: Saudi Arabia. Indian J of Human Gene. 17:207-211.

4. Akhtar MS, Qaw F, Borgio JF et al. (2013) Spectrum of α -Thalassemia Mutations in Transfusion-Dependent β-Thalassemia Patients from the Eastern Province of Saudi Arabia. Hemoglobin. 37:65-73.

5. Borgio JF, Abdulazeez S, Almandil NB et al. (2018) The α3.7 deletion in α-globin genes increases the concentration of fetal hemoglobin and hemoglobin A2 in a Saudi Arabian population. Mol Med Rep. 17:1879-1884.

6. Eissa M, Patel AA, Farag S et al. (2018) Awareness and Attitude of University Students About Screening and Testing for Hemoglobinopathies: case Study of the Aseer Region, Saudi Arabia. Hemoglobin. 42:264-268.

7. Al Asoom LI, Alsuwat HS, Rafique N et al. (2019) Functional DNA variations associated with Saudi female with low VO2max: a pilot microarray study. Am J Transl Res. 11:3659-3670.

8. Bowie LJ, Reddy PL, Beck KR (1997) Alpha thalassemia and its impact on other clinical conditions. Clinics in Laboratory Med. 17:97-108.

9. Reddy PL, Bowie LJ, Jiang H (2000) Blood pressure changes after intravenous administration of cell-free hemoglobin A and hemoglobin H in the rat. Nitric Oxide. 4:139-146.

10. Liem RI, Chan C, Vu T-HT et al. (2017) Association among sickle cell trait, fitness, and cardiovascular risk factors in CARDIA. Blood. 129:723-728.

11. Marlin L, Connes P, Antoine-Jonville S et al. (2008) Cardiorespiratory responses during three repeated incremental exercise tests in sickle cell trait carriers. Eur J Appl Physiol. 102:181-187.

12. Freund H, Lonsdorfer J, Oyono-Enguéllé S et al. (1995) Lactate exchange and removal abilities in sickle cell trait carriers during and after incremental exercise. Int J Sports Med. 16:428-434.

13. Le Gallais D, Prefaut C, Mercier J et al. (1994) Sickle cell trait as a limiting factor for high-level performance in a semi-marathon. Int J Sports Med. 15:399-402.

14. Diaw M, Samb A, Diop S et al. (2014) Effects of hydration and water deprivation on blood viscosity during a soccer game in sickle cell trait carriers. Br J Sports Med. 48:326-331.

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